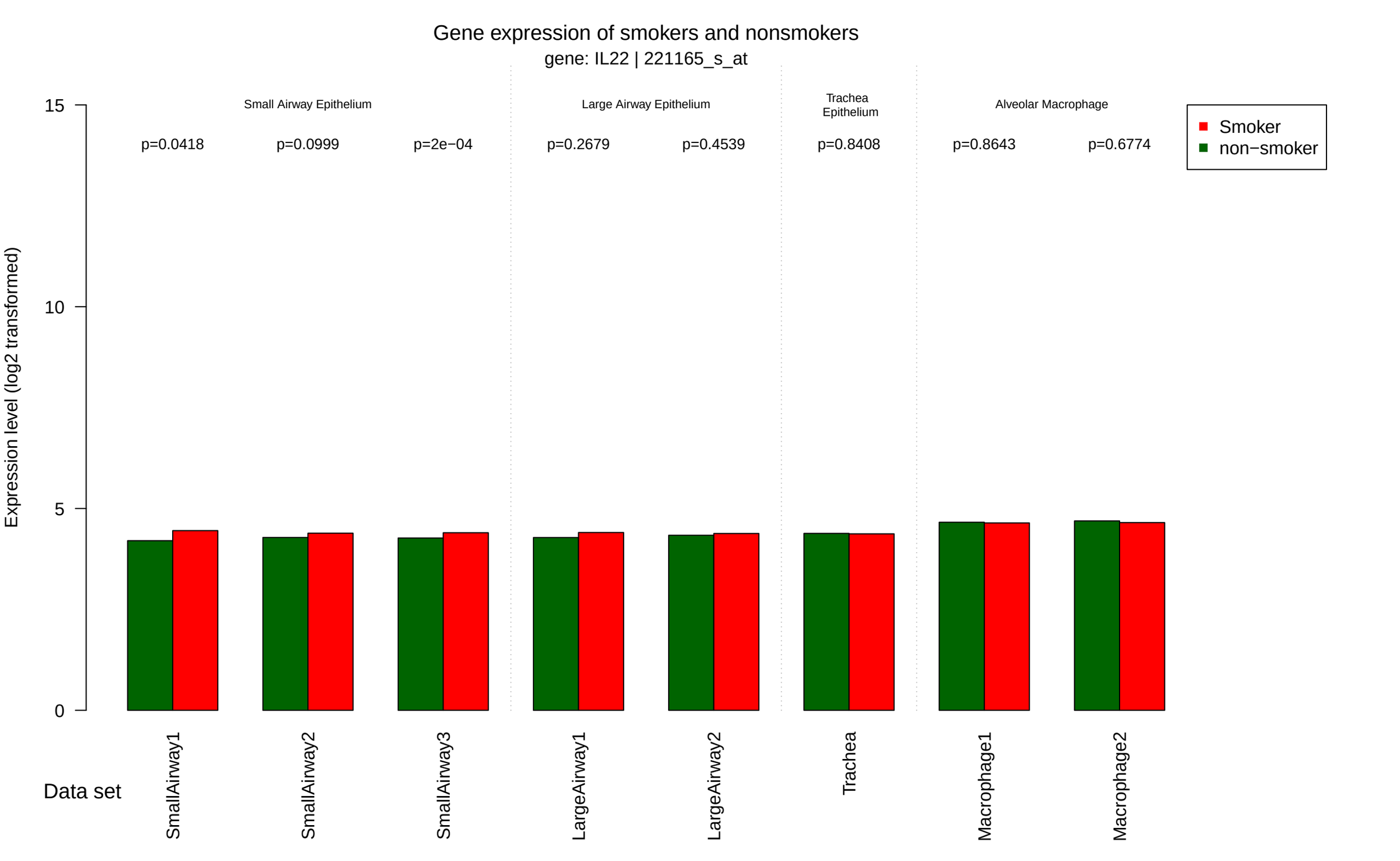
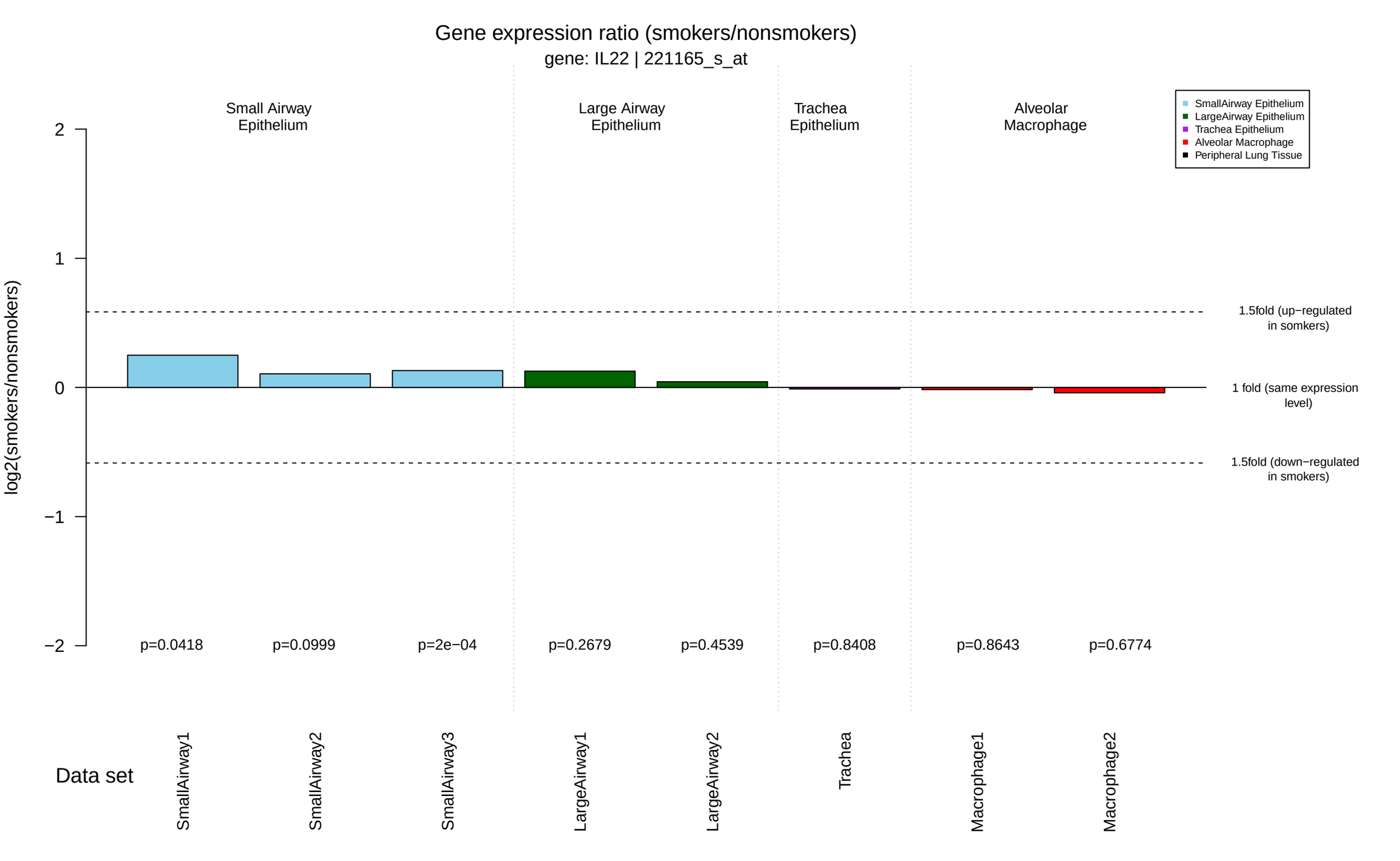
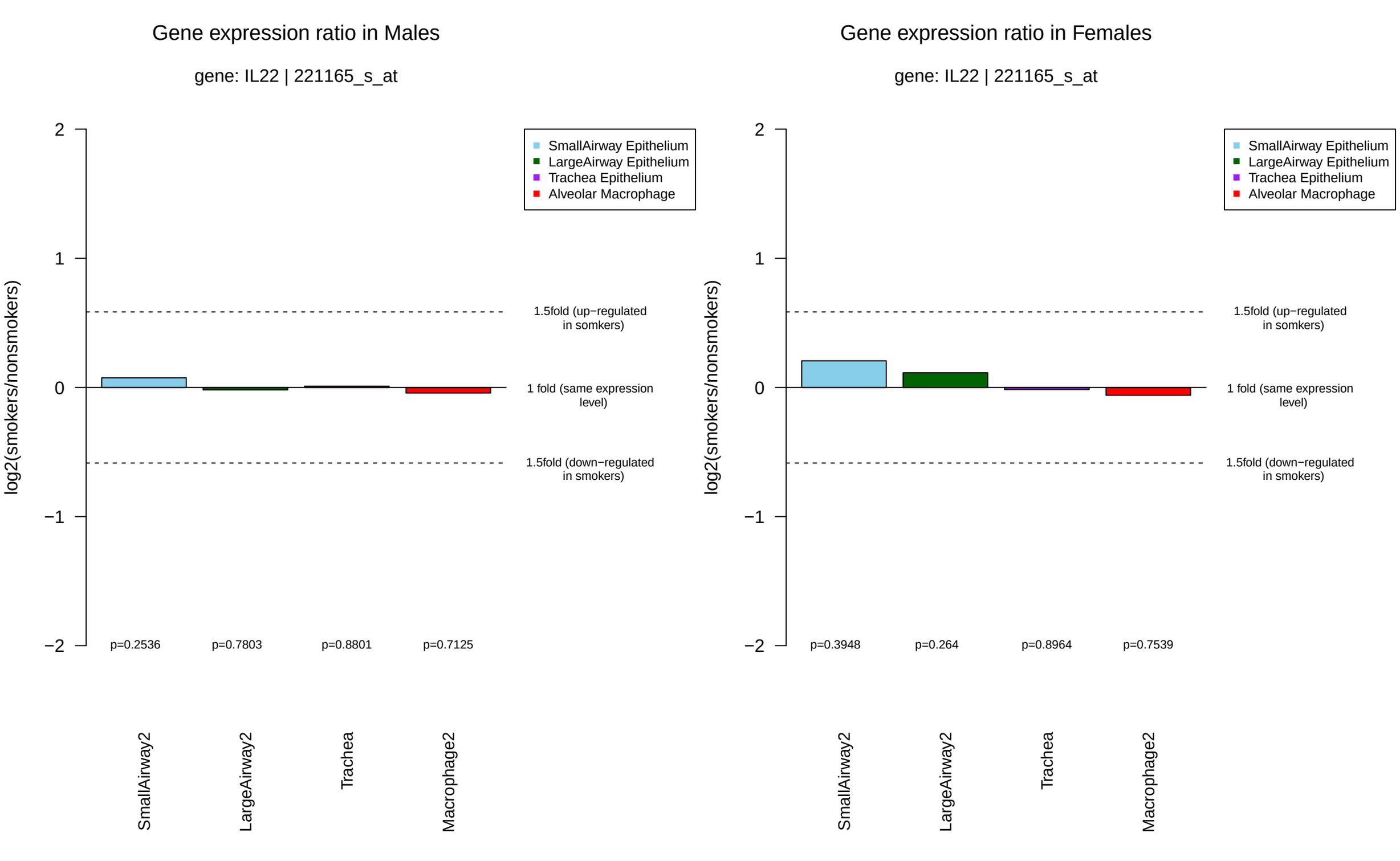
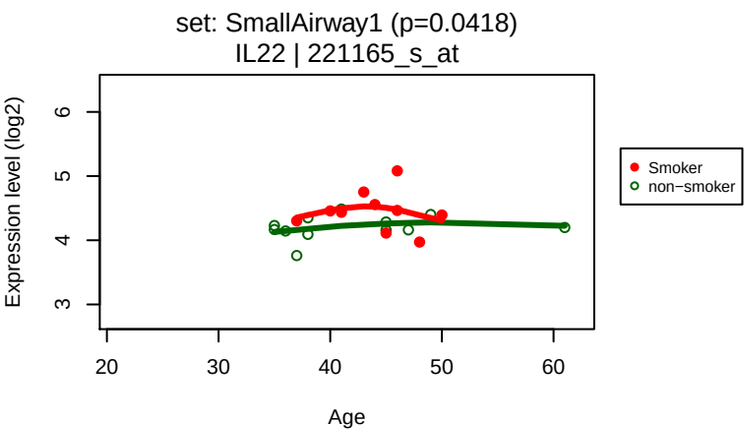
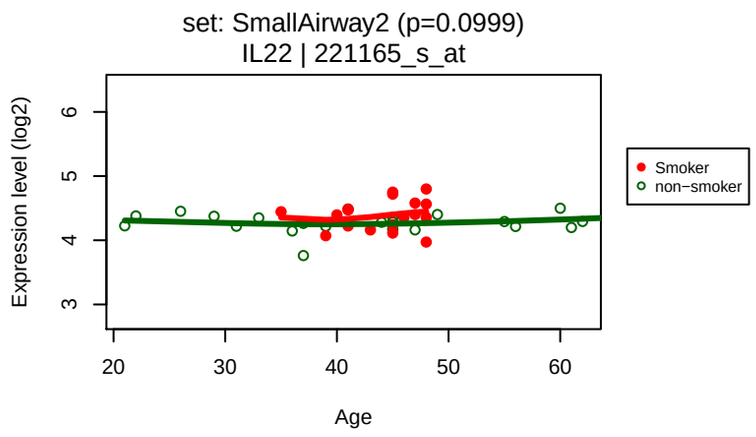
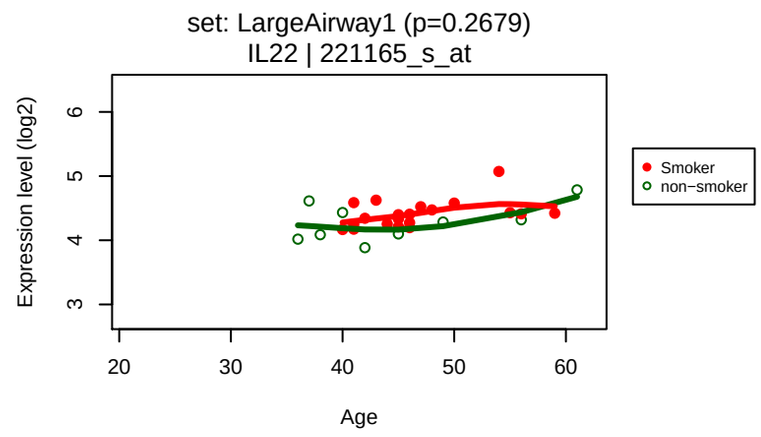

Gene expression of smokers and nonsmokers
gene: IL22 | 221165_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0418
p=0.0999
p=2e−04
p=0.2679
p=0.4539
p=0.8408
p=0.8643
p=0.6774
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

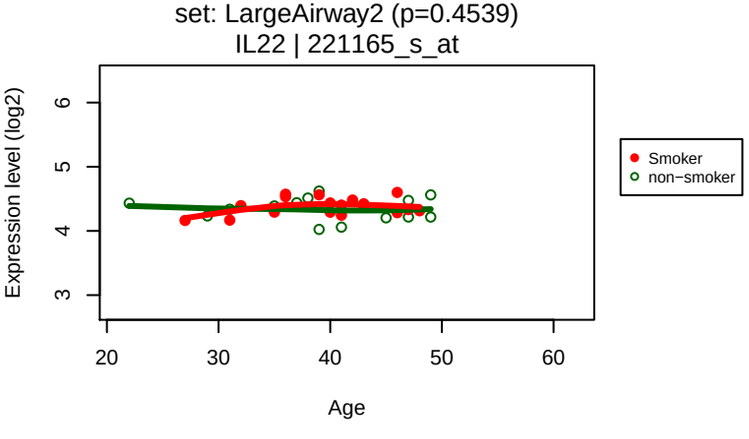

Gene expression ratio (smokers/nonsmokers)
gene: IL22 | 221165_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0418
p=0.0999
p=2e−04
p=0.2679
p=0.4539
p=0.8408
p=0.8643
p=0.6774
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

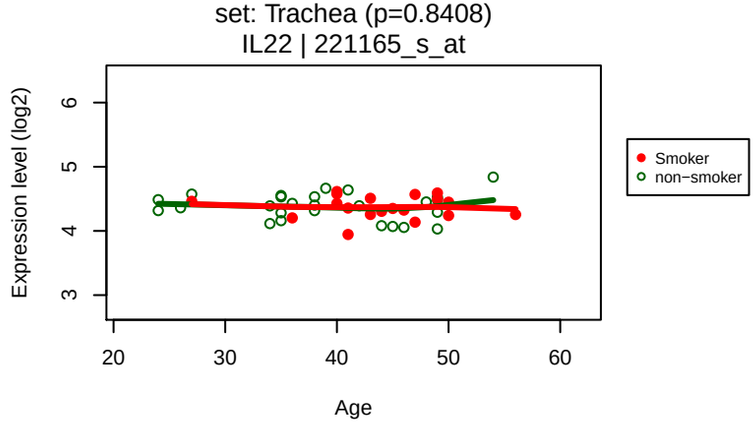

Gene expression ratio in Males
gene: IL22 | 221165_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2536
p=0.7803
p=0.8801
p=0.7125
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL22 | 221165_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3948
p=0.264
p=0.8964
p=0.7539
SmallAirway2
LargeAirway2
Trachea
Macrophage2

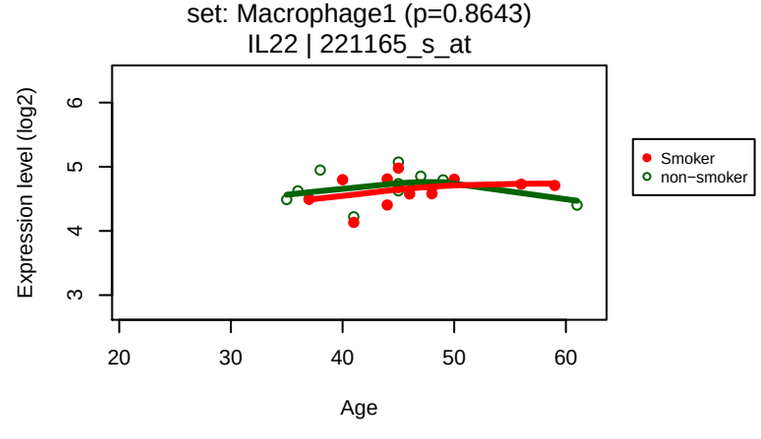

set: SmallAirway1 (p=0.0418)
IL22 | 221165_s_at
Expression level (log2)
Age
Smoker
non−smoker

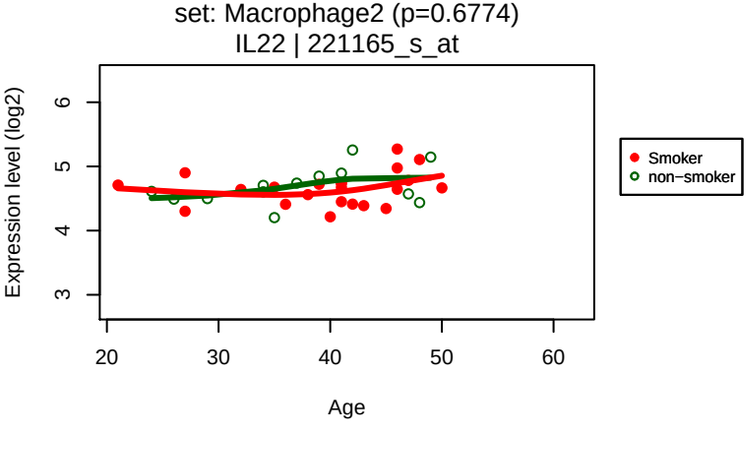

set: SmallAirway2 (p=0.0999)
IL22 | 221165_s_at
Expression level (log2)
Age
Smoker
non−smoker

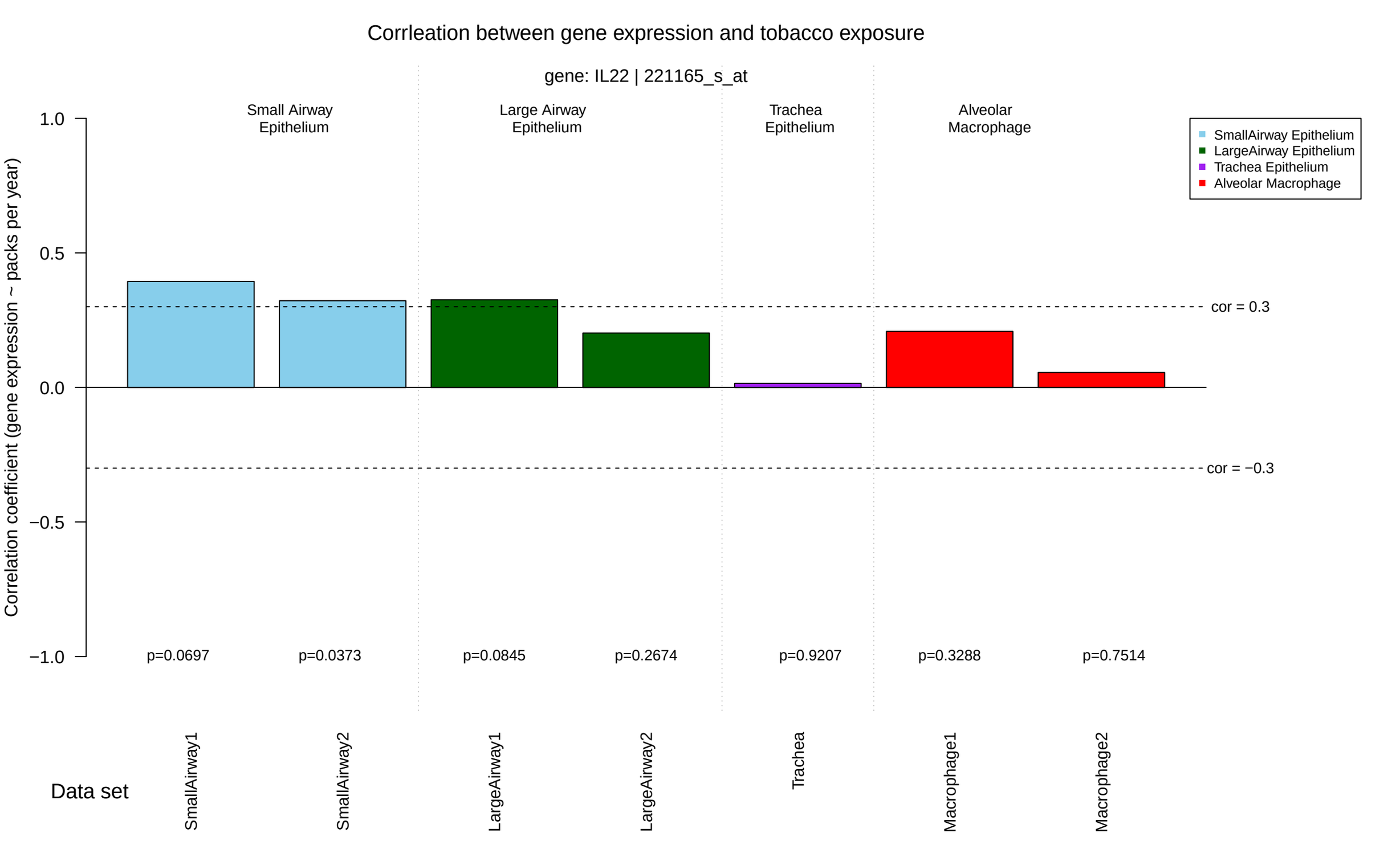

set: LargeAirway1 (p=0.2679)
IL22 | 221165_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4539)
IL22 | 221165_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8408)
IL22 | 221165_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8643)
IL22 | 221165_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6774)
IL22 | 221165_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL22 | 221165_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0697
p=0.0373
p=0.0845
p=0.2674
p=0.9207
p=0.3288
p=0.7514
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2